

Shortcut Learning Under Spurious Correlations

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Abstract—Deep neural networks have been observed to achieve high accuracy by exploiting spurious correlations instead of learning semantically meaningful features in the dataset; this phenomenon has been dubbed shortcut learning. In this project, we study whether a robustness-oriented training objective can reduce a model’s reliance on shortcut features for image classification applications. Using CIFAR-10 as our dataset, we synthetically introduced a class-correlated 3x3 monochrome patch into the top-left corner of a variable proportion of the training images to create a controlled shortcut signal. We trained the same convolutional neural network (CNN) under two objectives: Empirical Risk Minimization (ERM) and Group Distributionally Robust Optimization (GroupDRO). Both models were evaluated on biased, unbiased and anti-biased test sets to measure their capacity to generalize under distribution shifts. Saliency maps were also generated to allow for interpretation of the models’ attention. Results show that both ERM and GroupDRO achieve high accuracy when shortcuts are preserved, but their performance degrades substantially when the shortcut is removed in the unbiased case or reversed in the anti-biased case. GroupDRO consistently outperforms ERM on the unbiased test set and exhibits broader saliency over image content, indicating reduced but not eliminated shortcut reliance. Overall, GroupDRO improves robustness to spurious correlations, but neither objective was able to fully overcome shortcut learning under strong bias.

This project implementation can be found at:

<https://github.com/jlim0918/MIE424-group13-project-shortcut-learning/tree/main>

Index Terms—Shortcut Learning, Empirical Risk Minimization (ERM), Group Distributionally Robust Optimization (GroupDRO), Spurious Correlations, Image Classification, Robustness, Saliency maps.

1 INTRODUCTION

DEEP neural networks can produce high in-distribution accuracy even if the features that contribute to success are not semantically meaningful; this is referred to as “shortcut learning”. Shortcut Learning describes when a model learns from an easy and misleading correlation, rather than from the fundamental semantic structure of the problem [1]. As a result, if training and test data have an identical distributional bias, performance will remain high. In contrast, if the shortcut correlation is diminished, removed, or reversed, then performance will experience tremendous decline.

In this project, we explore shortcut learning in an image-classification context with controlled conditions, investigating whether a robustness-oriented optimization objective reduces shortcut reliance. We do this by training on the CIFAR-10 dataset and augmenting a variable proportion of the training images of each class with a monochrome patch corresponding to that class, applied to the upper-left hand corner of the image. This creates a biased training dataset such that the shortcut is highly predictive of the ground-truth label. The two optimization tasks we compare are (1) Empirical Risk Minimization (ERM), which minimizes expected training loss, and (2) Group Distributionally Robust Optimization (GroupDRO), which seeks to minimize the worst-case performance for all groups [2].

To measure robustness, we tested the trained model on three datasets: (1) a biased test dataset that preserves the shortcut-label correlation; (2) an unbiased test dataset that does not contain the shortcut; and (3) an anti-biased test dataset, where the shortcut-label correlation is reversed. We

seek to determine the extent to which shortcut reliance is observed across shifting distributions, given each optimization task. We used saliency maps as a qualitative indicator to examine whether either model primarily focused on the synthetic patches in the image rather than the actual content of the images, and to provide additional evidence of the model’s classification capabilities.

2 RELATED WORK

2.1 The Effect of Shortcut Learning

According to Geirhos et al. [1], shortcut learning is one of the most significant weaknesses of state-of-the-art deep neural networks systems, where the learned decision rules are often optimized only by being semantically meaningless (or meaningful) and therefore can also lead to misinterpretations of the true statistical regularity in the inference of the data. This is relevant to significant real-time variability, where the statistical regularities realized during the training phase no longer hold during inference. Building on this work we have created a synthetic shortcut configuration over the CIFAR-10 image dataset that allows us to define the magnitude of spurious correlations and investigate the extent to which the spurious shortcut optimizes more than the true visual signal.

2.2 The Optimization of Robustness

Traditional empirical risk minimization (ERM) assumes that by minimizing average training loss we will achieve generalized models. The issue of group imbalance (the

model responds to spurious signals that are more readily available than true signals only in training data or in the distribution) raises issues with the empirical risk for subgroups. To address these group imbalances Sagawa et al. [2] propose a Group DRO approach, replacing average loss minimization with the worst-case group loss-based minimization (optimization) approach which places robustness on both the training objective and the use of appropriate regularization of training losses. As our research continues from this literature stream, we evaluate whether the use of worst-case group loss and the cumulative group effects can reduce shortcut learning through the use of a synthetic environment.

2.3 Assessment Datasets and Interpretability

The benchmark datasets Waterbirds and CelebA [3] have provided an empirical platform to assess spurious correlations because they have relationships that are “real world” (causal) vs. “non-real world” (non-causal) within a data set by the presence of a background object or by the presence of an attribute with respect to the label. Evaluating generalizations beyond average i.i.d accuracy shows that it may be difficult to differentiate between models with low performance against shifting distributions (i.e. on certain minority groups) and model performance that has high average accuracies across the entire dataset. While our experimental design is based on assessing the influence of shortcut strength and the ability to optimize for much better generalization through synthetic patches with corresponding correlation value controls, we have also included concepts identified in other research that explores the impact of non-causal signals and their implications for modern-day machine learning [4]. The use of saliency maps (i.e. heat maps) will provide qualitative evidence of model attention by showing whether the model focuses on the synthetic patch or the image content.

3 DATASETS

3.1 Dataset Description

The CIFAR-10 (Canadian Institute For Advanced Research) is a dataset of images frequently used for training Machine Learning algorithms [5]. It is composed of 32x32 coloured images with an allocation of 50 000 entries in the training set and 10 000 entries in the testing set. There are 10 class labels in this dataset: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck.

3.2 Data Preparation

The CIFAR-10 dataset was augmented with “shortcut” features, in order to assess the models’ reliance on spurious correlations. The spurious feature was kept consistent across all augmented images, consisting of a 3x3 monochrome patch located in the top-left corner of the image with the colour of the patch corresponding to a different class. See Table 1 for a breakdown of the class-colour assignments and Figure 1 for a 16-image sample of the training set.

TABLE 1: CIFAR-10 Class Labels and Corresponding Shortcut Patch Colors

Class Label	Patch Color
Airplane	Bright Red
Automobile	Bright Green
Bird	Bright Blue
Cat	Bright Yellow
Deer	Pink
Dog	Teal
Frog	Dark Red
Horse	Dark Green
Ship	Dark Blue
Truck	Dark Yellow

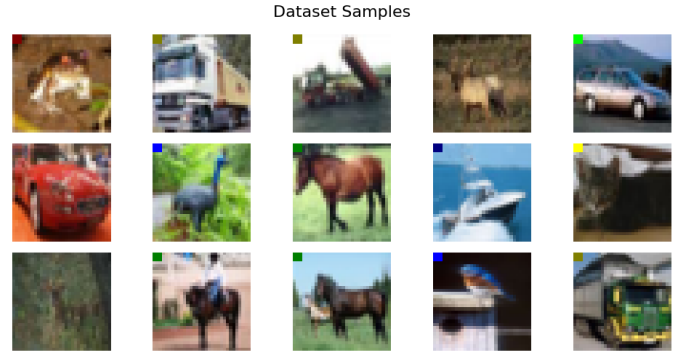


Fig. 1: 16-image sample of Augmented Training Set

To investigate spurious feature correlation, a variable augmentation ratio $p \in 0.0, 0.1, 0.2, \dots, 0.9$ was used with differing values between each run of the models. The shortcut feature was added onto $p\%$ of training samples ($p \times 50000$ images), while maintaining class balance, and the remaining $(1-p)\%$ ($(1 - p) \times 50000$ images) were left unmodified. For final test evaluation, the test dataset was duplicated and augmented to produce 3 separate sets of equal size: a non-biased set with no shortcut features, a biased set which follows the training set’s shortcut-class alignment, and an anti-biased set with shortcut features mismatched to the wrong class labels compared to the training set. See the figure below for a random sample of 16 images with spurious features.

3.3 Data Relevance

In the papers discussed within Related Work, the WaterBirds dataset was more commonly utilized with complex background features (e.g., water or land background) serving as the spurious feature. However, an augmented CIFAR-10 dataset was selected instead for this project, in order to compare how differing training objectives would perform on a more trivial, fixed-position spurious feature. Compared to the more variable background feature of Waterbirds, 3x3 patch in the modified CIFAR-10 better serves the purpose of visually showing, via saliency mapping, how the model(s) might be relying on the shortcut features.

4 METHODS

4.1 Model Architecture

The team designed a convolutional neural network for this image classification task. The architecture is currently composed of three convolutional layers for feature extraction

with a ReLU activation function and 2x2 max pooling in between each layer. These convolutional layers are then followed by two fully connected layers, ending in an output of 10 neurons, with one corresponding to each class. A dropout layer, with 0.2 dropout probability, is applied after the first fully connected layer to minimize overfitting.

4.2 Training Objectives

The previously mentioned model architecture will be trained under two separate objectives, ERM and GroupDRO, as outlined in the paper by Sagawa et al 2020 [2].

Empirical Risk Minimization:

$$\hat{\theta}_{\text{ERM}} := \arg \min_{\theta \in \Theta} \mathbb{E}_{(x,y) \sim \hat{P}} [\ell(\theta; (x, y))],$$

ERM is a common Machine Learning training objective, which seeks the set of model parameters θ from a model family Θ , which results in the minimum expected loss over the empirical distribution of the training dataset P .

GroupDRO:

$$\hat{\theta}_{\text{DRO}} := \arg \min_{\theta \in \Theta} \left\{ \hat{\mathcal{R}}(\theta) := \max_{g \in \mathcal{G}} \mathbb{E}_{(x,y) \sim \hat{P}_g} [\ell(\theta; (x, y))] \right\},$$

GroupDRO seeks the set of model parameters θ from a model family Θ , which minimizes the empirical worst-group (robust) risk R . This risk is defined as the maximum expected loss over the empirical distribution of the training set P_g , where g is a single group and $\mathcal{G}$ is the set of all groups in which the training data is divided. For the purpose of this project, groups will be allocated based on the presence of a spurious feature (boolean) and the class labels in the CIFAR-10 dataset, resulting in a total of $10 \times 2 = 20$ groups.

4.3 Optimization Algorithm

Adaptive Moment Estimation (Adam) was selected as the optimizer for both training objectives, as it is one of the most common optimizer algorithms used for neural networks [6]. The primary feature of Adam optimizer is that it utilizes dynamic learning rates for each parameter by estimating the first and second moments of the gradients. See the equation below for how weights, w_t , are updated for Adam. $\hat{m}_t$ and $\hat{v}_t$ represent the first and second moments, respectively, while α describes learning rate and ϵ describes a small error adjustment value.

$$w_{t+1} = w_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

4.4 Time Complexity

The time complexity for a single training epoch of the CNN implementation with GroupDRO training objective and Adam optimizer can be seen below. A single convolutional layer's time complexity is dependent on the number of images, N , the height, H , and width, W , of the input image, number of input channels, C , numbers of kernels, K , and kernel size, F . For fully-connected layers, time

complexity is based on number of neurons, N_{neurons} , and number of outputs, M . Adam's weight update scales based on number of parameters, P , while GroupDRO's group weighting updates scale according to number of groups, G [7].

TABLE 2: Time Complexity Breakdown

Component	Operation	Time Complexity
Conv-Layer	Forward/Backward	$O(N \cdot H \cdot W \cdot C \cdot K \cdot F \cdot F')$
FC-Layer	Linear Transform	$O(N_{\text{neurons}} \cdot M)$
Adam	Weight Update	$O(P)$
Group-DRO	Group Weighting	$O(G)$

The convolutional layer operations are the dominating factor for the overall implementation's time complexity. The computational cost difference between the ERM and GroupDRO training objectives relatively insignificant, with there being only 20 groups in this implementation.

4.5 Other Properties

The optimization problem is non-convex and continuous, because the CNN architecture is composed of multiple layers followed by non-linear activation functions. Adam optimizer mitigates the issue of non-convexity through its dynamically shifting step sizes. This optimizer is especially useful for GroupDRO, which is more subject to non-smooth error shifts due to adjusting parameter weights based on the maximum group loss.

5 EXPERIMENTS AND RESULTS

5.1 Experiment Setup

All experiments were run in Google Colab, utilizing T4 GPUs. The *tensorflow* package was used to extract and load the CIFAR-10 dataset, and *PyTorch* was utilized for creating the neural network. Both the ERM and GroupDRO models used the identical CNN architecture described in Section 4.1 and were trained under identical settings. Both models used the Adam optimizer with a learning rate of $1 \cdot 10^{-3}$, a batch size of 128, and a weight decay of $1 \cdot 10^{-4}$. To see how shortcut learning is impacted by the number of training data, we also varied the augmentation from 0% to 90% with increments of 10%. For each level, we evaluated the models on the biased, unbiased, anti-biased sets. For GroupDRO, groups were defined based on both the class label and the presence of a patch, resulting in 20 groups.

5.2 Training Accuracy

Both ERM and GroupDRO achieved near-perfect training accuracy across all augmentation ratios, reaching 90% or higher. This suggests that neither model underfits the training data, indicating that both ERM and GroupDRO are capable of fitting into the training data. Thus, possible differences in test accuracy is likely due to the training objectives, not the models themselves.

However, as the augmentation ratio is increased there is a slight dip in training accuracy. This might be due to the fact that since the majority of the data actually contains the color patch, identifying the remaining clean data might be harder for the model to learn.

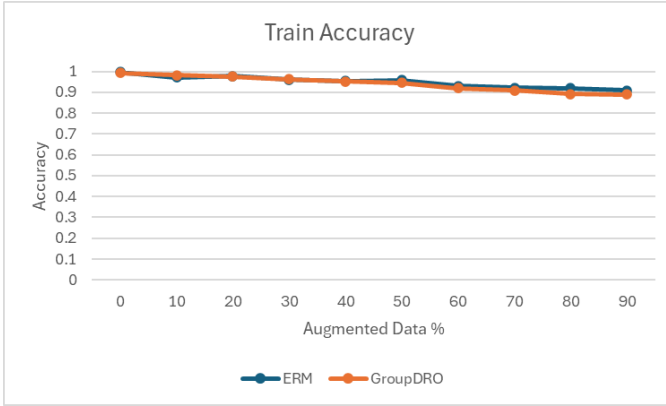


Fig. 2: Training accuracy for models against the percentage of augmented training data

5.3 Test Accuracy Across Datasets

On the biased set, both models quickly reached near-perfect accuracy starting from 40% augmentation. This is expected as the test set reflects the training set as both have the color patch so relying on the shortcut is a highly effective strategy. Interestingly, GroupDRO achieves a higher accuracy earlier than ERM so it may have picked up the spurious correlation quicker.

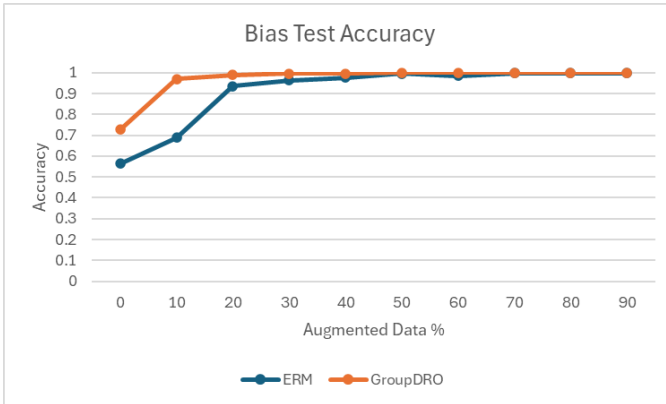


Fig. 3: Test accuracy of models against the percentage of biased data

On the unbiased set, we are able to see whether the models learned any meaningful visual features of the images. From the start at 0% augmentation, ERM achieves 55% accuracy while GroupDRO starts at around 70%. As augmentation ratio increases, the performance of both models decrease. ERM reaches around 35% accuracy while GroupDRO is at 52% at 90% augmentation. Throughout all the augmentation percentages, GroupDRO maintains a better accuracy of around 15 percentage points.

The gap shows that due to GroupDRO's focus on the worst performing group, it retains more attention to the image as a whole even if the spurious correlation is still the dominant force. However, as performance degrades for both models, it is clear that neither of them avoided shortcut learning. Another finding is that for ERM, a steeper decline happens going from 40% to 50% augmentation while a similar drop

happens to GroupDRO going from 60% to 70%. This delay shows that GroupDRO might be more robust to the augmentations.

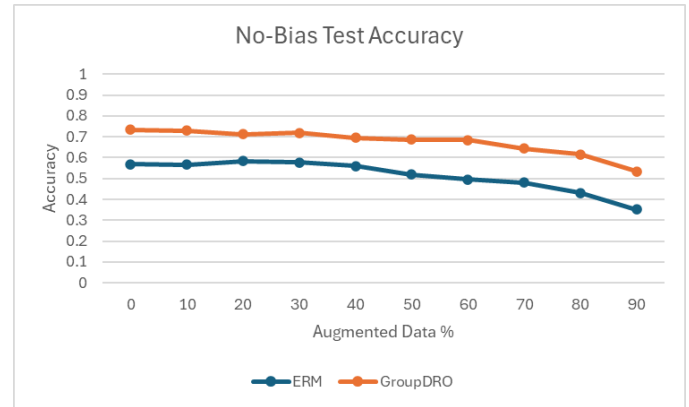


Fig. 4: Test accuracy of models against the percentage of non-biased data

On the anti-biased set, it is clearer to see how much the models depended on the shortcut learning as a model that mostly learned meaningful features should not be affected or affected minimally when the spurious correlations are modified.

However, both models show a steep drop in accuracy as augmentation increases. Both models reach near 0% accuracy at around 50% augmentation, which indicates that both models relied heavily on the colored patch to make predictions. Furthermore, for GroupDRO, just going from 0% to 10% augmentation was enough to drop accuracy from 70% to 20% accuracy, while for ERM, the case was going from 55% to 40%.

This difference of initial changes mirror the rise of accuracy on the biased test set, where the results indicated that GroupDRO may have picked up the spurious relation quicker. From 50% to 90% augmentation, both models are at effectively 0% accuracy which shows that both models have essentially started using the color patch itself to make the predictions.

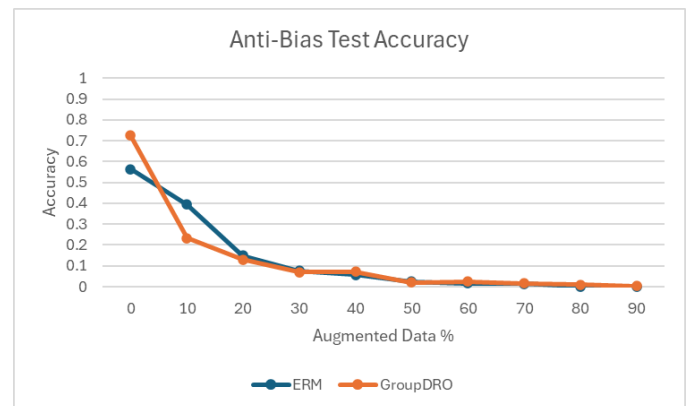


Fig. 5: Test accuracy of models against the percentage of anti-biased data

5.4 Interpretability: Saliency Maps

To better understand how the models made their predictions, we computed the saliency maps for each class using gradient-based attribution on the unbiased test set.

The saliency maps for ERM reveal the same patterns: across all classes, there is a strong and concentrated hotspot on the top left corner where the colored patch was located and nearly no activation anywhere else. This indicates ERM has effectively learned to rely on the colored patch only, which aligns with the results from before.

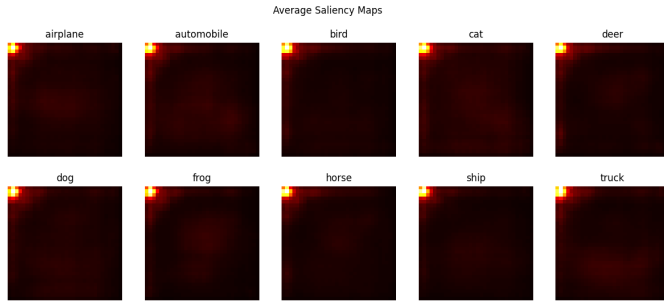


Fig. 6: Average saliency map for ERM

In contrast, the saliency maps for GroupDRO is more distributed. While the top left corner is still the most bright part, showing that the model still relied on the colored patch, there is also more activation for the rest of the image as well. This explains why GroupDRO outperformed on the clean test set. It was also able to learn some meaningful information to make predictions even when a colored patch was not present. However, one exception is the case of the frog class, where the saliency maps resembles one of ERM.

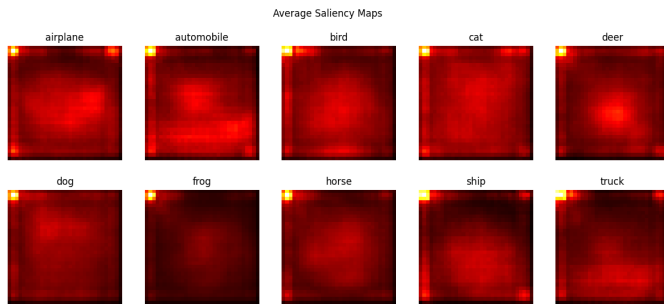


Fig. 7: Average saliency map for GroupDRO

Using the saliency maps, it appears that ERM uses a pure shortcut strategy, opting to rely on the colored patch to make predictions, while GroupDRO looks at the image in a slightly more balanced way, although the colored patch still dominates to some degree. For reference, you can look at the saliency maps for both models at 0% augmentation below, showing where both models looked initially without any color patches.

For no augmentation, both models show no signs of a concentration at the top left corner, and the attention to images are more broad. Both models look at similar regions for the same classes so the difference at higher augmentation levels are likely due to training objectives themselves.

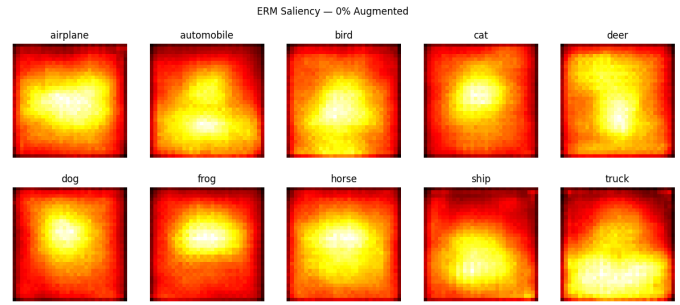


Fig. 8: Saliency maps for ERM at 0% augmentation

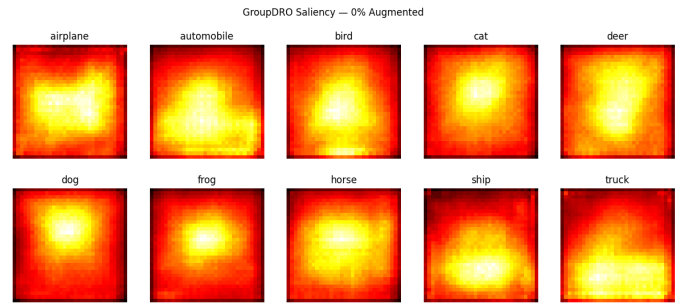


Fig. 9: Saliency maps for GroupDRO at 0% augmentation

6 CONCLUSION / FUTURE WORK

This project investigated shortcut learning in image classification using a controlled synthetic bias on the CIFAR-10 dataset. By introducing a class-correlated corner patch and comparing ERM against GroupDRO training objectives, we evaluated how each objective affected the robustness of the model under the distribution shift. The results show that both models perform well when the shortcut-label relationship is preserved, but both experience performance drops when the shortcut is removed or reversed. This confirms that both objectives learn to rely on the shortcut feature to some extent. However, GroupDRO maintains a higher accuracy on the unbiased test set which is further corroborated by the broader attention span observed in its saliency maps, suggesting it preserves more semantically meaningful visual features than ERM.

Despite this improvement, GroupDRO does not eliminate shortcut learning, as anti-biased accuracy still collapses under stronger shortcut biases. Future work could extend this study by testing stronger model architectures, alternative robust training methods, or different shortcut types such as variable-position patches or background-based spurious features. Additional experiments on datasets such as Waterbirds or CelebA would also help determine whether the observed trends generalize beyond this controlled CIFAR-10 setting.

7 END SECTIONS

7.1 Acknowledgements

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7.2 Contributions of team members

RS = research, FD = first draft, ET = Edit for grammar and spelling, CM = compile elements into the complete document, WC = write code

Task	Leen G.	Joaquin L.	Emre C.	Nic D.
Literature Review	RS, FD, ET, CM	RS, CM		
Project Proposal	FD, ET	FD, ET	FD, ET	FD, ET
Data Preprocessing	CM, WC	CM, WC		
Model Architecture and Training Functions		CM, WC	WC (Group-DRO)	
Hyperparameter Tuning		CM, WC	WC (Group-DRO)	
Presentation	RS, FD, ET, CM	RS, FD, ET, CM	RS, FD, ET, CM	ET, CM
Final Report				
Abstract		ET		FD, ET, CM
1. Introduction	RS, FD, ET, CM	ET		
2. Related Work	RS, FD, ET, CM			
3. Datasets		RS, FD, ET, CM		
4. Methods		RS, FD, ET, CM		
5. Experiments		ET	FD, ET, CM	
6. Conclusion				FD, ET, CM
7. End Sections		FD, CM, ET		
All Sections	CM, ET	CM, ET	CM, ET	CM, ET

7.3 References

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2023/file/0506ad3d1bcc8398a920db9340f27fe4-Paper-Conference.pdf

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